Database Management System

The big idea

A **database** is an organised collection of related data, like a school's record of students, marks and fees. A **DBMS (Database Management System)** is software to create, store, update and retrieve that data efficiently. You will use **LibreOffice Base**, which stores a whole database in one **.odb** file using its built-in **HSQLDB** engine.

Data, information and DBMS

- **Data** = raw facts (e.g. 85, "Riya"). **Information** = processed, meaningful data ("Riya scored 85 in IT").
- Examples of DBMS: MySQL, Oracle, Microsoft Access, LibreOffice Base, PostgreSQL.
- **Advantages**: less **data redundancy** (duplication), less **inconsistency**, data sharing, data security, data integrity, backup and recovery.

Relational database terms

In a **relational database (RDBMS)** data is stored in **tables (relations)**.

Term	Meaning
Field / attribute / column	One piece of information, e.g. RollNo, Name
Record / tuple / row	All information about one item, e.g. one student

Term	Meaning
Primary key	Field that uniquely identifies each record; cannot be empty or repeated (e.g. AdmissionNo)
Composite key	Primary key made of two or more fields together
Candidate key	Any field(s) that could serve as primary key
Alternate key	A candidate key not chosen as primary key
Foreign key	Field in one table that refers to the primary key of another table, linking them

NAME IS A POOR PRIMARY KEY

Two students can share a name. Choose a field that is always unique, like admission number.

Data types in Base

- **Numeric**: Boolean (Yes/No), Tinyint, Smallint, Integer, BigInt, Numeric, Decimal, Real, Float, Double.
- **Alphanumeric/text**: Text [VARCHAR], Text (fix) [CHAR], Memo [LONGVARCHAR].
- **Date/Time**: Date, Time, Date/Time (Timestamp).
- **Binary**: Image [LONGVARBINARY] for photos.

Creating a database and table

1. **File > New > Database** → *Create a new database* (Embedded HSQLDB) → Next → *Yes, register the database* → Finish → name it (e.g. School.odb) → Save.
2. In **Tables**, click **Create Table in Design View**.

3. Type **Field Names**, choose **Field Type**, set **Field Properties** (length, default, required).
4. Right-click the row selector of the unique field → **Primary Key**.
5. Save (Ctrl+S) and name the table. Double-click the table to enter records (**Datasheet view**).

You can also use **Use Wizard to Create Table**. Sort with the Sort buttons and filter with AutoFilter in datasheet view.

Relationships

Tools > Relationships – add tables and drag a primary key to the matching foreign key.

- **One-to-one** – one record matches exactly one (student ↔ ID card).
- **One-to-many** – one record matches many (one class has many students). Most common.
- **Many-to-many** – many match many (students ↔ clubs), handled through a linking table.
  Relationships maintain **referential integrity** (you cannot enter a foreign key value that has no matching primary key).

Queries

A **query** asks a question of the database and shows matching records. In Base, under **Queries**: **Create Query in Design View**, **Use Wizard to Create Query**, or **Create Query in SQL View**.

SQL QUERY

```
SELECT Name, Marks FROM Student WHERE Marks > 80 ORDER BY Marks DESC;
```

Shows names and marks of students scoring above 80, highest first. In

Design View you would add the table, tick the fields, type `>80` in the **Criterion** row and choose *descending* in **Sort**.

SQL commands: **DDL** (CREATE, ALTER, DROP – define structure) and **DML** (SELECT, INSERT, UPDATE, DELETE – work with data).

Forms and reports

- **Form** – a friendly screen for entering and viewing data one record at a time. **Forms > Use Wizard to Create Form**: choose table, fields, arrangement (columnar, tabular…), style; finish.
- **Report** – formatted, printable output of data, often with grouping and totals. **Reports > Use Wizard to Create Report**: choose fields, labels, grouping, sort order, layout. Reports can be **static** or **dynamic** (updated each time they open).

FORM VS REPORT

Form = **input** (and view). Report = **output** for printing/summary.

Quick recap

- DBMS reduces redundancy and inconsistency; Base files are .odb.
- Table = rows (records) × columns (fields). Primary key is unique and not null; foreign key links tables.
- Tools > Relationships: 1:1, 1:many, many:many.
- Queries retrieve data (Design view / Wizard / SQL). Forms enter data; reports present it.

Practice questions

1. The default file extension of a LibreOffice Base database is
   (a) .ods (b) .odt (c) .odb (d) .dbf
2. A row in a table is also called a
   (a) record or tuple (b) field or attribute (c) key (d) query
3. Which field is the best choice for a primary key in a Student table?
   (a) Name (b) City (c) AdmissionNo (d) DateOfBirth
4. A field in one table that refers to the primary key of another table is a
   (a) candidate key (b) composite key (c) alternate key (d) foreign key
5. The relationship "one class has many students" is
   (a) one-to-one (b) one-to-many (c) many-to-many (d) many-to-one only
6. Which SQL command retrieves data from a table?
   (a) INSERT (b) UPDATE (c) SELECT (d) CREATE
7. Which Base object is mainly used to present data in a printable format?
   (a) Table (b) Form (c) Query (d) Report
8. Differentiate between a primary key and a foreign key.
9. Write an SQL query to display Name and Marks of all students from table Student whose Marks are more than 75, in descending order of Marks. Also name the two other ways of creating a query in Base.

Answers: 1. c 2. a 3. c 4. d 5. b 6. c 7. d 8. see notes 9. see notes